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Tiered Risk Assessment for Petroleum Hydrocarbons C6–C9: A Case Study at a Typical Decommissioned Petroleum Refinery Site in Gansu Province

Kaixuan Zhu, Chao Zhang, Guanlin Guo * and Rongxiao Yuan

Technical Centre for Soil, Agriculture and Rural Ecology and Environment, Ministry of Ecology and Environment, Beijing 100012, China; zhukaixuan@tcare-mee.cn (K.Z.)
* Correspondence: guoguanlin@tcare-mee.cn

Abstract

No method to assess the risks of petroleum hydrocarbon pollutants C6–C9 in soils on construction land in China has been established. At one decommissioned petroleum refinery site in northwestern China, we performed an innovative tier 3 risk assessment method using carbon fraction proportions. Using HJ 25.3 guidelines, the risk-screening value for soil contamination of land by petroleum hydrocarbons was 192 mg kg^{-1} for industrial land use. However, based on site-specific parameters, this value was 226 mg kg^{-1}, with a corresponding contaminated soil volume of 381,904 m^3. A tier 3 risk assessment incorporating carbon fraction proportions and site-specific parameters yielded a risk control value of 2370 mg kg^{-1} and reduced the soil volume requiring decontamination to 87,047 m^3, potentially saving CNY 324 million (~USD 45.5 million as of November 2025) in remediation costs. Therefore, implementing a tier 3 risk assessment for C6–C9 pollutants can optimize remediation strategies and enhance the precision and scientific rigor of petroleum hydrocarbon-contaminated soil remediation.

Keywords: petroleum hydrocarbons; risk assessment; soil; contaminated site; remediation

1. Introduction

China is the world's largest oil-refining producer [1]. Because petroleum hydrocarbon (PHC) contamination of soils and groundwater (from emissions, seepage, and accidents during petroleum refining) can have serious implications [2–5], an understanding of the distribution and environmental risks posed by PHCs in soil at petrochemical-contaminated sites is important [5,6].

PHCs are a complex mix of compounds that primarily comprise alkanes (e.g., linear and branched alkanes, cycloalkanes, alkenes, and alkynes), aromatics (e.g., mono- and polycyclic aromatic hydrocarbons), asphaltenes (e.g., phenols, fatty acids, ketones, and porphyrins), and resins (e.g., pyridine, quinoline, carbazoles, sulfonates, and amides) [3,7]. Compared with analyses of single pollutants, the pronounced differences in toxicity and physicochemical properties of PHC constituents, along with practical challenges in laboratory-based quantification of individual components [8], complicate the risk assessment process [9,10]. Currently, human health risk assessment methods for PHCs can be classified into indicator, total petroleum hydrocarbon, and/or fractionation approaches.

In practice, these methods are often implemented in combination based on specific assessment requirements [9]. For example, in the United States (US), a combination of each

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method is used [11], whereas in Australia [12] and Canada [13], a combination of two of them (indicator and fractionation) is used. The Chinese standard "Soil environmental quality risk control standard for soil contamination of development land" (GB36600-2018) [14] uses a method that integrates indicator and fractionation methods. The indicator method analyzes five benzene homologs and eight polycyclic aromatic hydrocarbons, and the fractionation method analyzes PHCs C10–C40 (but not C6–C9). Despite the promulgation of standard analytical methods for PHCs C6–C9 and C10–C40 in soils and sediments in 2019 [8], no established assessment criteria and risk assessment studies have targeted hydrocarbons C6–C9 in China.

Short-chain PHCs present non-negligible environmental risks because of constituents beyond typically monitored indicators such as benzene, toluene, ethylbenzene, and xylene (BTEX). In the US Environmental Protection Agency (USEPA) Office of Underground Storage Tanks "Petroleum Hydrocarbon Vapor Intrusion" database, the total PHC-to-benzene ratio exceeds 900:1 in 33% of samples, suggesting that the vapor intrusion risk is primarily attributable to aliphatic hydrocarbons. Controlling benzene concentrations below screening levels alone cannot sufficiently manage the non-carcinogenic risks posed by volatile PHCs [15]. Industries using petroleum products as raw materials (e.g., those producing coatings, fragrances, rubber, and plastics) primarily emit low-carbon-chain aromatic hydrocarbons as major pollutants [16]. In the Pearl River Delta, short-chain volatile PHCs are the dominant soil contaminants, and USEPA risk assessment methods have been used to characterize their non-carcinogenic health risks [17]. From the perspectives of screening values and land-use types, Jiang et al. demonstrated the need to perform refined risk assessments for complex multi-component contaminants [18]. Implementing a tier 3 risk assessment at large volatile organic compound (VOC)-contaminated sites can also substantially reduce remediation costs, yielding significant economic benefits [19].

We characterize the distribution and risk of PHCs C6–C9 in soils at a former petroleum refinery site in northwestern China. Based on systematic sampling and analysis, we describe the spatial distribution and human health risks posed by these PHCs and evaluate the feasibility and benefits of a tiered risk assessment framework. In doing so, we provide a scientific foundation for the risk-based management and remediation of petrochemical-contaminated sites.

2. Materials and Methods

2.1. Study Area

The study area, formerly occupied by a decommissioned refinery on the Longdong Loess Plateau, Qingyang City, Gansu Province, covers a total area of ~21,060 m^2 (Figure 1). The refinery was operational from 1970 until 2004 and produced various petroleum products (civilian liquefied petroleum gas, gasoline, kerosene, diesel, and solvent oil). The site was functionally divided into refining and chemical zones; an eastern section contained storage tanks for fuels such as gasoline and diesel; the central area was dedicated to technology development and laboratory facilities; a northwestern zone housed crude oil storage tanks and catalytic cracking units; and a southwestern area contained wastewater treatment facilities and loading/unloading stations for petroleum products.

Based on soil stratigraphic structure and hydrogeological characteristics, we divided the soil profile into four layers for sampling. Layer (a), 0–3.0 m: miscellaneous fill from the Holocene series of the Quaternary period (Q4ml); layer (b), 3.0–6.5 m: silty soil of alluvial–pluvial origin from the Holocene series of the Quaternary period (Q4al + pl); layer (c), 6.5–9.5 m: rounded gravel of alluvial–pluvial origin from the Holocene series of the Quaternary period (Q4al + pl); and layer (d), 9.5–15 m: Neogene mudstone (N). The

maximum depth of the miscellaneous fill layer was 3.0 m, and the average depth to the groundwater table was ~8.5 m.

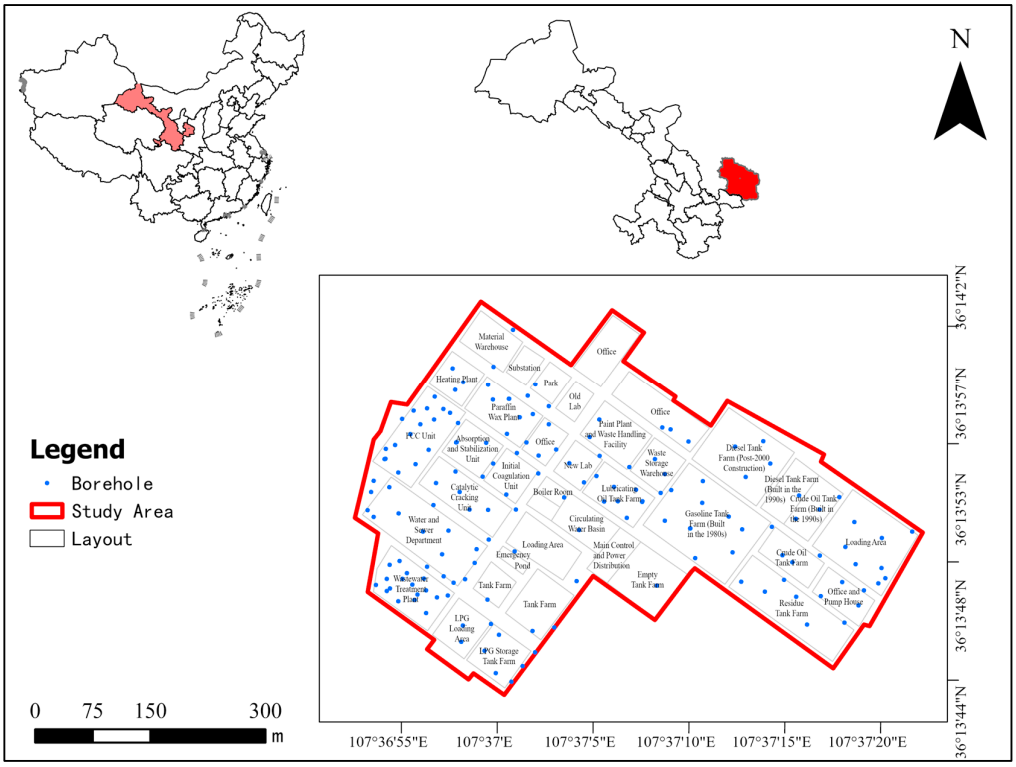

Figure 1. Location of the study area.

2.2. Sample Collection and Testing

Following the Technical Guidelines for Soil Pollution Status Investigation of Land for Construction (HJ 25.1-2019) [20] and Technical Guidelines for Monitoring of Soil Pollution Risk Control and Remediation for Land for Construction (HJ 25.2-2019) [21], a grid-based sampling method was used to identify potential migration and dispersion pathways across the study area, combined with a judgmental source-tracking approach based on reconnaissance survey data to pinpoint potential contamination sources. Soils were sampled in 20 m × 20 m grids, with one sampling point per grid, with additional sampling points then concentrated in areas of high contamination potential (e.g., the wastewater treatment zone, production workshops, and storage tank areas). During field sampling, a photoionization detector was used for rapid on-site screening at each potential sampling location. Sampling decisions were guided by these real-time screening results combined with professional assessment of field conditions. A total of 139 sampling points (Figure 1) were established, and 259 soil samples were collected.

2.2.1. Soil Sample Collection and Preservation

Soil samples were collected using an SH-30 drill rig. (Wuxi Tongda Geoligical Prospecting Machinery Factory, Jiangsu Province, China) Undisturbed soil samples were obtained with a stainless-steel shovel. After removing the surface soil with a scraper, samples from target depths were collected and immediately transferred into 250 mL wide-mouth sampling bottles. Bottles were filled completely, sealed without headspace, and stored at 4 °C in darkness.

2.2.2. Sample Testing and Laboratory Quality Control

PHCs C6–C9 in soil samples were analyzed according to guidelines in Soil and Sediment—Determination of Petroleum Hydrocarbons (C6–C9)—Purge and Trap/Gas Chromatography (HJ 1020-2019) [22]. Benzene homologs were determined according to guidelines in Soil and Sediment—Determination of Volatile Organic Compounds—Purge and Trap/Gas Chromatography–Mass Spectrometry (HJ 605-2011) [23]. The testing methods and equipment are detailed in Table 1.

Table 1. Testing methods and equipment.

Parameter	Analytical Method	Limit of Detection	Instrumentation	Relative Percent Difference/(%)
Petroleum hydrocarbons C6–C9	HJ 1020-2019	0.04 mg·kg^{-1}	Purge and trap gas Chromatography–mass spectrometry Amoxt 7890/5977B	0.7–11.3
Benzene	HJ 605-2011	1.9 μgkg^{-1}		0.3–14.0
Toluene		1.3 μgkg^{-1}		1.1–19.8
Ethylbenzene		1.2 μgkg^{-1}		0.4–9.7
Xylene		1.2 μgkg^{-1}		0.7–13.4

2.3. Risk Assessment Models and Parameters

Conceptual exposure model layering incorporates multiple factors, including geological stratification and future development depth. Taking into consideration both the stratigraphic conditions revealed during the investigation and potential future groundwater development scenarios, we characterized soils into surface (0–6.5 m) and subsurface (6.5–15.0 m) soils when calculating risk-screening values of contamination in development land.

We focused on assessing the non-carcinogenic risks of PHCs C6–C9 via six exposure pathways: (1) oral ingestion of soil (ois); (2) dermal contact with soil (dcs); inhalation of (3) soil particles (pis); (4) outdoor air containing gaseous contaminants derived from surface soil (iov1); (5) outdoor air containing gaseous contaminants derived from subsurface soil (iov2); and (6) indoor air containing gaseous contaminants derived from subsurface soil (iiv).

The human health risk assessment model is represented by Equations (1)–(21) [24].

(1) Soil risk control value (SRCV) based on non-carcinogenic effects via oral ingestion, $HCVS_{ois}$

$$HCVS_{ois} = RfD_o \times SAF \times AHQ \times \frac{BW_a \times AT_{nc}}{OISER_a \times ED_a \times EF_a \times ABS_o \times 10^{-6}} \quad (1)$$

(2) SRCV based on non-carcinogenic effects via dermal contact, $HCVS_{dcs}$

$$HCVS_{dcs} = \frac{RfD_o \times ABS_{gi} \times SAF \times AHQ \times BW_a \times AT_{nc}}{239 \times H_a^{0.417} \times BW_a^{0.517} \times SER_a \times SSAR_a \times EF_a \times ED_a \times E_v \times ABS_d \times 10^{-6}} \quad (2)$$

(3) SRCV based on non-carcinogenic effects via inhalation of soil particles, $HCVS_{pis}$

$$HCVS_{pis} = \frac{\frac{RfC \times DAIR_a}{BW_a} \times RfD_i \times SAF \times AHQ \times BW_a \times AT_{nc}}{PM_{10} \times DAIR_a \times ED_a \times PIAF \times (fspo \times EFO_a + fspi \times EFI_a) \times 10^{-6}} \quad (3)$$

(4) SRCV based on non-carcinogenic effects via inhaling outdoor air containing gaseous contaminants derived from surface soil, $HCVS_{iov1}$

$$HCVS_{iov1} = \frac{RfC \times DAIR_a}{BW_a} \times SAF \times AHQ \times \frac{BW_a \times AT_{nc}}{VF_{suroa} \times DAIR_a \times EFO_a \times ED_a} \quad (4)$$

(5) SRCV based on non-carcinogenic effects via inhaling outdoor air containing gaseous contaminants derived from subsurface soil, $HCVS_{iov2}$

$$HCVS_{iov2} = \frac{RfC \times DAIR_a}{BW_a} \times SAF \times AHQ \times \frac{BW_a \times AT_{nc}}{VF_{suboa} \times DAIR_a \times EFO_a \times ED_a} \quad (5)$$

(6) SRCV based on non-carcinogenic effects via inhaling indoor air containing gaseous contaminants derived from subsurface soil, $HCVS_{iiv}$

$$HCVS_{iiv} = \frac{RfC \times DAIR_a}{BW_a} \times SAF \times AHQ \times \frac{BW_a \times AT_{nc}}{VF_{subia} \times DAIR_a \times EFI_a \times ED_a} \quad (6)$$

(7) Formulae for the volatilization factor of contaminant diffusion (FVFCD) from surface soil to outdoor air:

$$VF_{suroa} = MIN(VF_{suroa1}, VF_{suroa2}) \quad (7)$$

$$VF_{suroa1} = \frac{\rho_b}{DF_{oa}} \times \sqrt{\frac{4 \times D_s^{eff} \times H'}{\pi \times \tau \times 3,153,600 \times K_{sw} \times \rho_b}} \times 1000 \quad (8)$$

$$DF_{oa} = \frac{U_{air} \times W \times \delta_{air}}{A} \quad (9)$$

$$D_s^{eff} = D_a \times \frac{\left(1 - \frac{\rho_b}{\rho_s} - \frac{\rho_b \times \rho_{ws}}{\rho_w}\right)^{3.33}}{\left(1 - \frac{\rho_b}{\rho_s}\right)^2} + D_w \times \frac{\left(\frac{\rho_b \times \rho_{ws}}{\rho_w}\right)^{3.33}}{H' \times \left(1 - \frac{\rho_b}{\rho_s}\right)^2} \quad (10)$$

$$K_{sw} = \frac{\rho_{ws}}{\rho_w} + K_{oc} \times \frac{f_{om}}{1700} + H' \times \left(\frac{1}{\rho_b} - \frac{1}{\rho_s} - \frac{\rho_{ws}}{\rho_w}\right) \quad (11)$$

$$VF_{suroa2} = \frac{d \times \rho_b}{DF_{oa} \times \tau \times 3,153,600} \times 10^3 \quad (12)$$

(8) FVFCD from subsurface soil to outdoor air:

$$VF_{suboa} = MIN(VF_{suboa1}, VF_{suboa2}) \quad (13)$$

$$VF_{suboa1} = \frac{1000}{\left(1 + \frac{DF_{oa} \times L_s}{D_s^{eff}}\right) \times \frac{K_{sw}}{H'}} \quad (14)$$

$$VF_{suroa2} = \frac{d_{sub} \times \rho_b}{DF_{oa} \times \tau \times 3,153,600} \times 10^3 \quad (15)$$

(9) FVFCD from subsurface soil to indoor air:

$$VF_{subia} = MIN(VF_{subia1}, VF_{subia2}) \quad (16)$$

$$VF_{suboa1} = \frac{1000}{\left(1 + \frac{D_s^{eff}}{DF_{ia} \times L_s} + \frac{D_s^{eff} \times L_{crack}}{D_{crack}^{eff} \times L_s \times \eta}\right) \times \frac{K_{sw}}{H'}} \quad (17)$$

$$DF_{ia} = L_B \times ER \times \frac{1}{86,400} \quad (18)$$

$$D_{cracl}^{eff} = D_a \times \frac{\theta_{acrack}^{3.33}}{(\theta_{acrack} + \theta_{wcrack})^2} + D_w \times \frac{\theta_{wcrack}^{3.33}}{H' \times (\theta_{acrack} + \theta_{wcrack})^2} \quad (19)$$

$$VF_{suria2} = \frac{d_{sub} \times \rho_b}{DF_{ia} \times \tau \times 3,153,600} \times 10^3 \quad (20)$$

(10) SRCV based on the integrated non-carcinogenic effects of all six exposure pathways, $HCVS_n$:

$$HQ_n = \frac{1}{\frac{1}{HCVS_{ois}} + \frac{1}{HCVS_{dcs}} + \frac{1}{HCVS_{pis}} + \frac{1}{HCVS_{iov1}} + \frac{1}{HCVS_{iov2}} + \frac{1}{HCVS_{iiv}}} \quad (21)$$

The parameters in the assessment model and their corresponding values are presented in Table 2.

Table 2. Exposure parameters and assigned values.

Parameter	Definition	Data			
		Tier 1	Source	Tier 2	Source
$OISER_a$	Daily oral ingestion rate of soils of adults, mg·d^{-1}	100	[24]	100	[24]
ED_a	Exposure duration of adults, a	25	[24]	25	[24]
EF_a	Exposure frequency of adults, d·a^{-1}	250	[24]	250	[24]
ABS_o	Absorption factor of oral ingestion (unitless)	1	[24]	1	[24]
BW_a	Average body weight of adults, kg	61.8	[24]	58.4	[25]
AT_{nc}	Average time for non-carcinogenic effect, d	9125	[24]	9125	[24]
SAF	Soil allocation factor (unitless)	0.33	[24]	0.33	[24]
AHQ	Acceptable hazard quotient for individual contaminant (unitless)	1	[24]	1	[24]
H_a	Average height of adults, cm	161.5	[24]	161.5	[24]
SER_a	Skin exposure ratio of adults (unitless)	0.18	[24]	0.18	[24]
$SSAR_a$	Adherence rate of soil on skin for adults, mg·cm^{-2}	0.2	[24]	0.2	[24]
E_v	Daily exposure frequency of dermal contact event, once·d^{-1}	1	[24]	1	[24]
PM_{10}	Content of inhalable particulates in ambient air, mg·m^{-3}	119	[24]	52	[26]
$DAIR_a$	Daily air inhalation rate of adults, m3·d^{-1}	14.5	[24]	14.6	[25]
$PIAF$	Retention fraction of inhaled particulates in body (unitless)	0.75	[24]	0.75	[24]
$fspo$	Fraction of soil-borne particulates in outdoor air (unitless)	0.5	[24]	0.5	[24]
EFO_a	Outdoor exposure frequency of adults, d·a^{-1}	62.5	[24]	62.5	[24]
$fspi$	Fraction of soil-borne particulates in indoor air (unitless)	0.8	[24]	0.8	[24]
EFI_a	Indoor exposure frequency of adults, d·a^{-1}	187.5	[24]	187.5	[24]
ρ_b	Soil bulk density, kg·dm^{-3}	1.5	[24]	1.71	Measured
τ	Averaging time for vapor flux, a	25	[24]	25	[24]
U_{air}	Ambient air velocity in mixing zone, cm·s^{-1}	200	[24]	200	[24]
W	Width of source zone area, cm	4000	[24]	55,827	Measured
δ_{air}	Mixing zone height, cm	200	[24]	200	[24]
A	Source zone area, cm^2	16,000,000	[24]	982,160,000	Measured
ρ_s	Density of soil particulates, kg·dm^{-3}	2.65	[24]	2.71	Measured
ρ_{ws}	Soil water content, kg·kg^{-1}	0.2	[24]	0.162	Measured
ρ_w	Density of water, kg·dm^{-3}	1	[24]	1	[24]
f_{om}	Organic matter content in soils, g·kg^{-1}	15	[24]	15	[24]
d	Concentrations of contaminants in surface soil, cm	50	[24]	650	Measured
L_s	Concentrations of contaminants in subsurface soil, cm	50	[24]	650	Measured
d_{sub}	Thickness of subsurface soil, cm	100	[24]	850	Measured
L_B	Volume/infiltration area ratio of enclosed space, cm	300	[24]	300	[24]
ER	Air exchange rate of enclosed space, once·d^{-1}	20	[24]	20	[24]
η	Areal fraction of cracks in foundations/walls (unitless)	0.005	[24]	0.005	[24]
L_{crack}	Thickness of enclosed-space foundation or wall, cm	35	[24]	35	[24]
θ_{acrack}	Soil air content—soil-filled foundation cracks (unitless)	0.26	[24]	0.26	[24]
θ_{wcrack}	Soil water content—soil-filled foundation cracks (unitless)	0.12	[24]	0.12	[24]
H'	Henry's law constant (unitless)	1.86	[27]	1.86	[27]
D_a	Diffusivity in air, cm^2·s^{-1}	7.35×10^{-2}	[27]	7.35×10^{-2}	[27]
D_w	Diffusivity in water, cm^2·s^{-1}	8.38×10^{-6}	[27]	8.38×10^{-6}	[27]
K_{oc}	Organic carbon partition coefficient, cm^3·g^{-1}	1.72×10^2	[27]	1.72×10^2	[27]
RfD_o	Chronic oral reference dose, mg·kg^{-1}·d^{-1};	0.005	[28]	0.005	[28]

Table 2. *Cont.*

Parameter	Definition	Data			
		Tier 1	Source	Tier 2	Source
RfC	Chronic inhalation reference concentration, mg·cm^{-3}	0.4	[28]	0.4	[28]
ABS_{gi}	Fraction of contaminant absorbed in gastrointestinal tract (unitless)	1	[28]	1	[28]
ABS_d	Fraction of contaminant absorbed dermally from soil (unitless)	/	[28]	/	[28]

Note: "/" denotes "not applicable".

2.4. Tiered Risk Assessment

Because there is always an element of uncertainty in risk assessment [29], the USEPA introduced a three-tiered risk assessment approach in its 1992 exposure assessment guidelines. This methodology aims to derive more scientific and accurate risk assessment conclusions by constructing realistic exposure scenarios, optimizing exposure assessment models, and refining the selection of assessment parameters [30–32].

Tier 1 entails developing a conceptual site model based on conservative principles and using default parameters for assessment. This tier carries relatively high uncertainty and is typically used for preliminary risk screening. Tier 2 builds upon this by incorporating site-specific parameters (e.g., soil physicochemical properties and hydrogeological conditions) and substituting measured data into the assessment model to enhance reliability. Tier 3 further refines the assessment by integrating the transport and transformation behavior of contaminants in the environment; by making detailed adjustments to assessment models and toxicity parameters, it improves the scientific validity and accuracy of risk assessment conclusions.

We performed a tier 1 risk assessment with reference to the Technical Guidelines for Risk Assessment of Soil Contamination in Construction Land (HJ 25.3-2019) [24], using recommended values for short-chain aliphatic hydrocarbons from the USEPA Regional Screening Level Summary Table (2024.11) [27,28]. For Tier 2 risk assessment, certain assessment parameters were refined using site-specific measured values (Table 1). For the Tier 3 risk assessment, toxicity parameters for contaminants were referenced from the Canada-wide standard for PHCs in soil user guidance [13], specifically the recommended values for the F1 (C6–C10) fraction. A weighted calculation based on the proportional composition of each sub-fraction was performed to further optimize risk assessment results.

$$HCVS_i = \frac{1}{\sum \frac{MF_{subfraction_j}}{HCVS_{subfraction_j}}} \tag{22}$$

where i represents different exposure pathways (ois, dcs, pis, iov1, iov2, and iiv); $MF_{subfraction_j}$ represents the mass fraction of each subfraction within F1 (Table 3); and $subfraction_j$ represents aliphatic hydrocarbons C6–C8, aliphatic hydrocarbons C > 8 to C10, and aromatic hydrocarbons C > 8 to C10.

Table 3. Subfraction and parameter values of F1 in Canada.

Parameter	Definition	Data			
		Aliphatic Hydrocarbons C6–C8	Aliphatic Hydrocarbons C > 8 to C10	Aromatic Hydrocarbons C > 8 to C10	Source
H'	Henry's law constant (unitless)	50	80	0.48	[33]
D_a	Diffusivity in air, cm^2·s^{-1}	0.1	0.1	0.1	[33]

Table 3. *Cont.*

Parameter	Definition	Data			Source
		Aliphatic Hydrocarbons C6–C8	Aliphatic Hydrocarbons C > 8 to C10	Aromatic Hydrocarbons C > 8 to C10	
D_w	Diffusivity in water, $cm^2 \cdot s^{-1}$	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	[33]
K_{oc}	Organic carbon partition coefficient, $cm^3 \cdot g^{-1}$	3.98×10^3	3.16×10^4	1.58×10^3	[33]
RfD_o	Chronic oral reference dose, $mg \cdot kg^{-1} \cdot d^{-1}$;	5	0.1	0.04	[33]
RfC	Chronic inhalation reference concentration, $mg \cdot cm^{-3}$	18.4	1	0.2	[33]
ABS_{gi}	Fraction of contaminant absorbed in gastrointestinal tract (unitless)	1	1	1	[34]
ABS_d	Fraction of contaminant absorbed dermally from soil	0.2	0.2	0.2	[34]
$MF_{subfraction}$	Mass fraction of each subfraction within the F1 (unitless)	0.55	0.36	0.09	[34]

3. Results

3.1. Characteristics of PHCs C6–C9 Contamination in Soil

Of the 259 soil samples, 171 were contaminated with PHCs C6–C9 (a detection rate of 66.0%). The peak concentration was 5870 mg kg^{-1}, with percentile values of 17.45 mg kg^{-1} (75th), 301.8 mg/kg (90th), and 3757 mg kg^{-1} (99th). Based on analytical results and stratigraphic structure, contamination distribution maps of PHCs C6–C9 were generated for soil layers (Figure 2).

The distribution of PHCs C6–C9 in soils is closely related to the historical production layout of the industrial facility. Highly contaminated areas were primarily concentrated in the former paraffin production area, catalytic cracking unit, storage tank zone, wastewater treatment plant, and loading/unloading platforms. This distribution pattern aligns with findings from other petroleum refinery sites, where VOC contamination is typically focused in tank farms, wastewater treatment areas, production units, and loading platforms [35]. This consistency suggests a commonality in the spatial distribution of contamination at petroleum refinery sites, which can inform the identification of key areas and the design of sampling strategies for similar sites.

Vertically, contaminant migration is influenced by multiple factors, such as adsorption, soil moisture content, temperature, and the physicochemical properties of the contaminants [36,37]. Only one high-concentration point was detected in surface soils (0–3.0 m). This differs from some other studies [38,39], possibly because of the extended idle period following site shutdown and accelerated volatilization and degradation of PHCs C6–C9 in surface soils caused by anthropogenic disturbances (e.g., removal of facilities and equipment). Notably, over 50% of the high-concentration points for PHCs C6–C9 occurred within 6.5–9.5 m depth, near where groundwater levels fluctuate between high- and low-water periods. This accumulation is likely because of the physicochemical properties of PHCs C6–C9, which facilitate their accumulation at the capillary fringe and potential formation of a light non-aqueous phase liquid (LNAPL) [40]. This makes this depth interval a high-risk zone for contaminant accumulation.

Correlation analysis was performed using 259 datasets. A significant positive correlation existed between PHCs C6–C9 and m&p-xylene ($R = 0.85$; $p < 0.01$), suggesting potential common sources or similar environmental behaviors (Figure 3). Areas with elevated m&p-xylene concentrations in petroleum-refining sites may warrant attention for potential co-contamination by PHCs C6–C9. Correlations between PHCs C6–C9 and benzene, while

statistically significant, were extremely weak (R = 0.16; $p < 0.01$), indicating that these contaminants have distinct sources or differing migration and transformation processes.

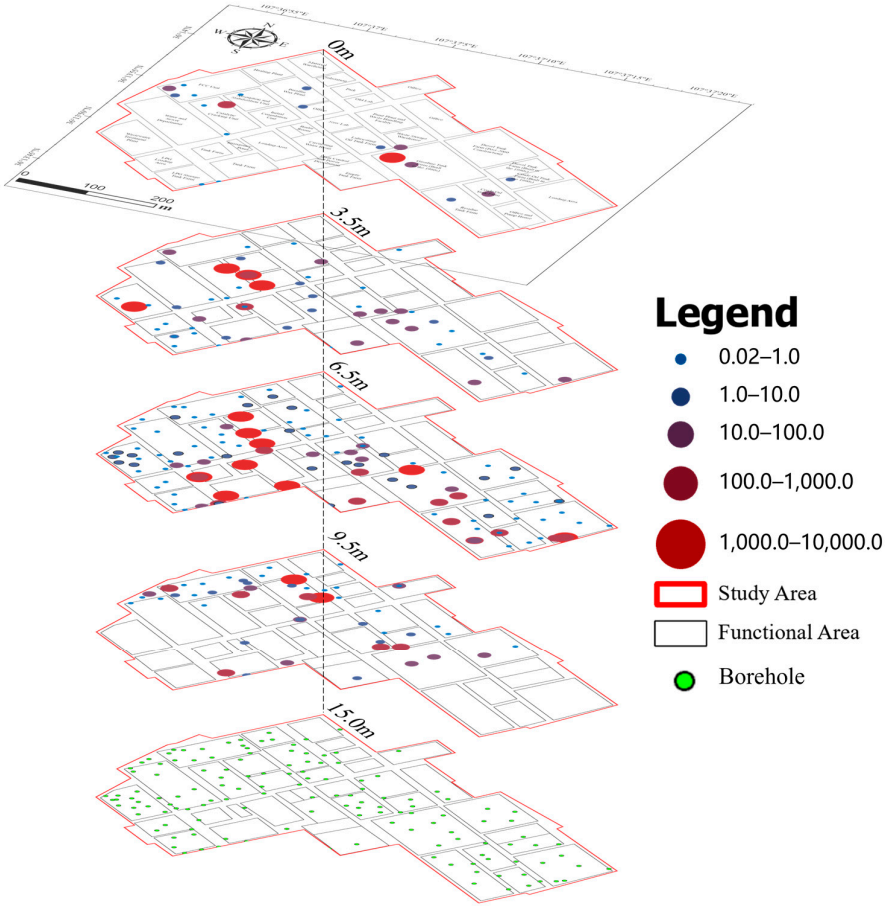

Figure 2. Distribution of petroleum hydrocarbons (PHCs) C6–C9 in soil by depth.

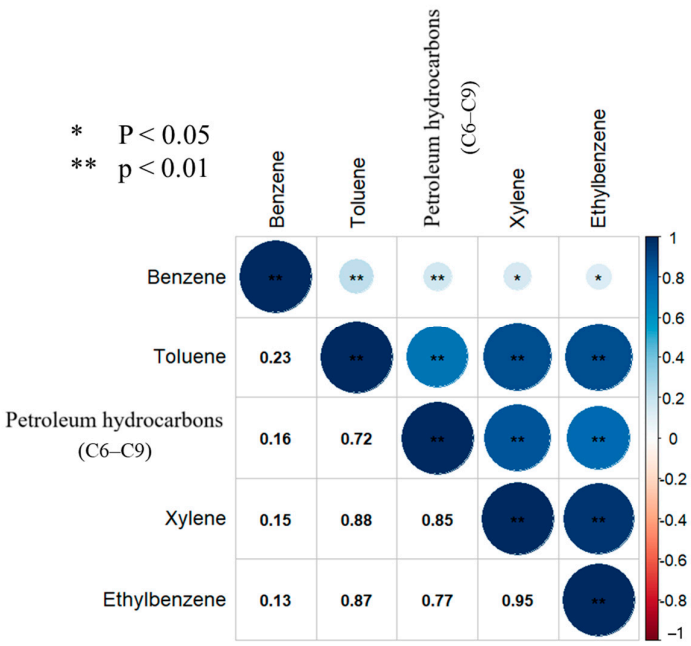

Figure 3. Correlation of PHCs C6–C9 and benzene, toluene, ethylbenzene, and xylene (BTEX) concentrations in soil.

3.2. Risk Assessment Results

3.2.1. Tier 1

Based on the recommended parameters from the "Technical Guidelines for Risk Assessment of Soil Contamination on Land for Construction" (HJ 25.3-2019), we performed a tier 1 risk assessment, deriving a tier 2 land-use screening value of 192 mg kg^{-1} for PHCs C6–C9. To evaluate the reasonableness of this screening value, we compared it against standards for PHCs C6–C9 under similar land-use scenarios for various countries (Table 4). Significant disparities existed. New Zealand's soil-screening standard (120 mg kg^{-1}) was the most stringent, applicable to sandy-gravel surface soils from a 0 to 4.0 m depth. In contrast, Australia's health-screening value (26,000 mg kg^{-1}) was the most lenient, applicable to scenarios involving direct contact but excluding the vapor intrusion pathway. These differences primarily stem from variations in the risk assessment models used, the depth of toxicological understanding of PHCs C6–C9, and the regional characteristics of key exposure parameters. The screening value derived using China's current HJ 25.3 risk assessment model falls within a relatively stringent range, reflecting a precautionary principle in risk management that aims to protect human health to the greatest possible extent.

Table 4. Screening values for PHCs C6–C9 in soils by country.

No.	Country/Region	Standard Name	Target Compound/Fraction	Value (mg kg^{-1})	Source
1	Hong Kong, China	Risk-Based Remediation Goals	C6–C8	Industry/park 10,000	[41]
2	USA	Regional Screening Level	Total PHCs (aliphatic low)	Industry 1900	[28]
3	Canada	Canada-Wide Standard for PHC in Soil User Guidance	F1	Commercial/industry 170–240 Soil texture-specific	[34]
4	New Zealand	Guidelines for Assessing and Managing PHC-Contaminated Sites in New Zealand Module 4 Tier 1 Soil Acceptance Criteria	C7–C9	Commercial/industry 120–16,000 Soil texture- and depth-specific	[42]
5	Australia	Health Screening Levels for PHC in Soil and Groundwater Summary	C6–C10	Commercial/industry 310–26,000 Soil texture, depth, and land-use specific	[12]

Sun et al. demonstrated that toxicity assessment and risk characterization can lead to significant differences in soil-screening values [43]. Variations in toxicity assessment among fractions largely explain differences in PHCs C6–C9 soil-screening values between countries or regions. Additionally, the risk characterization methods used in Canada differ from those in the US, Hong Kong, and elsewhere, possibly further contributing to the divergence between Canada's soil-screening values and those elsewhere.

3.2.2. Tier 2 vs. 3

Based on site-specific parameters, the Tier 2 risk assessment derived a risk control value of 226 mg kg^{-1} for PHCs C6–C9. Further incorporating the proportional contributions of sub-fractions through weighted calculation, the tier 3 risk assessment established a significantly higher (2370 mg kg^{-1}) risk control value. Using these two values as benchmarks and integrating analytical data with field conditions, the spatial distribution of contamination was characterized using ArcGIS 10.4. This process delineated areas that exceeded calculated risk control values, calculated their areal extent, and quantified corresponding soil volumes (Figures 4 and 5). The key point of the risk control value derived

from the tier 3 risk assessment lies in adoption of Canada's $MF_{subfraction}$ values. This is also the main source of uncertainty in this method. For specific contaminated sites, the $MF_{subfraction}$ should be tested where feasible to reduce the uncertainty associated with this approach.

Figure 4. Spatial extent of contaminated soil under a tier 2 risk assessment. (**a**) 0–3.0 m area beyound Tier 2 risk control value; (**b**) 3.0–6.5 m area beyound Tier 2 risk control value; (**c**) 6.5–9.5 m area beyound Tier 2 risk control value; (**d**) 9.5–15.0 m area beyound Tier 2 risk control value.

Figure 5. Spatial extent of contaminated soil area under a tier 3 risk assessment. (**a**) 0–3.0 m area beyound Tier 2 risk control value; (**b**) 3.0–6.5 m area beyound Tier 2 risk control value; (**c**) 6.5–9.5 m area beyound Tier 2 risk control value; (**d**) 9.5–15.0 m area beyound Tier 2 risk control value.

Contaminated soil volumes corresponding to each risk assessment tier are summarized in Table 5. For the Tier 2 assessment, the total contaminated soil volume (381,904 m^3) was mostly concentrated from a 6.5 to 9.5 m depth (238,096 m^3). In contrast, the total contaminated soil volume identified by the tier 3 assessment was significantly reduced to 87,047 m^3, with the majority located in shallower (0–6.5 m depth) soils.

Table 5. Comparison of contaminated soil volumes for different tiers of risk assessment.

Depth (m)	Tier 2 (m^3)	Tier 3 (m^3)	Reduction in Volume (m^3)	Reduction Rate (%)
0–3.0	17,734	17,734	0	0
3.0–6.5	38,407	38,407	0	0
6.5–9.5	238,096	30,906	207,190	87.0
9.5–15.0	87,668	0	87,668	100.0
Total	381,904	87,047	294,857	77.2

Performing a tier 3 risk assessment can significantly optimize the remediation scope. Within 0–3.0 m and 3.0–6.5 m depth ranges, the volume of contaminated soil was unchanged. However, from 6.5 to 9.5 m and 9.5 to 15.0 m, the contaminated soil volume decreased by 87.0% and 100.0%, respectively. This resulted in a total reduction of ~295,000 m^3, representing a 77.2% decrease, which signifies substantial environmental and economic benefits.

Significant differences exist in the content and toxicity of alkane and aromatic fractions of PHCs [44]. Current Chinese standards for risk control of PHCs C10–C40 are based on toxicity parameters of the relatively more toxic aromatic fraction (C10–C16). This leads to conservative risk assessment results with considerable uncertainty when using a single fraction approach [8]. While refined risk assessment methods based on carbon fraction hypotheses for PHCs C10–C40 have been established in Shanghai and Guangdong [45], nothing comparable exists for PHCs C6–C9 elsewhere in China. Drawing on relevant studies from Canada and the Total Petroleum Hydrocarbon Criteria Working Group, we pioneer a refined, fraction-based risk assessment for PHCs C6–C9. This provides methodological support for precise risk management of similarly contaminated sites and aligns with the risk-based "strict-before-lenient" screening principle [46].

3.3. Economic Benefits

For shallow areas (depths of 0–6.5 m) where excavation is feasible, ex situ remediation technology is used. Given the physical properties of PHCs C6–C9, ambient temperature desorption technology could be considered. For deeper areas (depths of 6.5–15.0 m) where excavation is less feasible, in situ remediation technology could be used. Given that PHCs C6–C9 do not contain benzene series components, which reduces the difficulty of ring-opening reactions, in situ chemical oxidation technology could be considered. The unit costs of these two technologies are 500 CNY m^{-3} and 1100 CNY m^{-3}, respectively. Based on feasible technologies and their cost analysis, combined with contaminated soil volumes determined by tiered risk assessment, we estimated the corresponding direct remediation costs (Table 6). Costs corresponding to tier 2 risk assessment amount to CNY 386 million (~USD 54.2 million as of November 2025), whereas those for tier 3 risk assessment are reduced to CNY 62 million (~USD 8.7 million as of Nov 2025) (an ~84% decrease of CNY 324 million (~USD 45.5 million as of November 2025)).

Table 6. Estimation of remediation costs under different tiers of risk assessment.

Tiered Risk Assessment	Remediation Technology	Cost USD m^{-3} (CNY as of November 2025)	Remediation Volume m^3	Total Cost million (CNY as of November 2025)
Tier 2	Ambient temperature desorption (0–6.5 m)	70.2 (500)	56,141	54.2 (386)
	In situ chemical oxidation (6.5–15.0 m)	154.47 (1100)	325,763	

Table 6. *Cont.*

Tiered Risk Assessment	Remediation Technology	Cost USD m^{-3} (CNY as of November 2025)	Remediation Volume m^3	Total Cost million (CNY as of November 2025)
Tier 3	Ambient temperature desorption (0–6.5 m)	70.2 (500)	56,141	8.7 (62)
	In situ chemical oxidation (6.5–15.0 m)	154.47 (1100)	30,906	

Our results demonstrate that performing a tier 3 risk assessment with higher precision for PHCs C6–C9-contaminated sites can accurately delineate the actual risk area and optimize the volume of soil requiring remediation, thereby yielding significant economic benefits. This approach provides a valuable reference for risk management and remediation decision making at similar contaminated sites.

4. Discussion

4.1. Tiered Risk Assessment Enhances Precision and Aligns with Green, Low-Carbon Principles

The complexity of PHC composition underscores the need for conducting refined risk assessments. Park et al. demonstrated that detailed quantitative risk assessment was needed to determine scientifically and economically appropriate cleanup target levels [47]. We established a method to assess the risks of PHCs C6–C9 in soils on construction land in China. Tier 3 risk assessment for PHCs C6–C9 can significantly reduce the remediation cost (an ~84% decrease of CNY 324 million (~USD 45.5 million as of November 2025)), aligning with China's green and low-carbon requirements for soil remediation (Table 6). Tiered risk assessment enables more accurate assessment, increases remediation goals by up to 80%, and substantially reduces remediation volume [7,48]. Both existing studies and our work confirm the feasibility of implementing tiered risk assessment for PHCs. In China, PHCs account for nearly 50% of organic-contaminated sites [49], further demonstrating the significant potential of this approach in practical applications.

4.2. Localized Studies Are Needed to Establish Additional Parameters for PHC Risk Assessment

To implement the risk-based soil environmental management strategy grounded in the three core elements of "source–pathway–receptor," Chinese researchers have performed localization studies on exposure parameters related to the receptor element [25,50]. While the properties of most contaminants in the source element are relatively stable, PHCs exhibit compositional diversity, structural heterogeneity, varied origins, and divergent environmental behaviors. Accordingly, an investigation into the localized characteristics of PHC pollution sources is required. Currently, technical guidelines for Guangzhou, China, provide risk assessment parameters for PHCs C10–C40 [45]. However, as in our study, these parameters are also derived from sources such as the US and Canada. Therefore, investigating compositional characteristic parameters of PHCs in contaminated sites across China represents a key direction for future work and a prerequisite for advancing tiered risk assessment of PHCs.

5. Conclusions

In petroleum refinery-contaminated sites, monoaromatic hydrocarbons do not fully indicate the contamination risks of short-chain PHCs. We recommend that PHCs C6–C9 be included as key characteristic contaminants in contamination identification and incorporated into soil contamination status investigations and risk assessment procedures.

At our study site, a tier 3 risk assessment based on carbon fraction assumptions for PHCs C6–C9 determined a required remediation volume of 87,047 m^3, with corresponding remediation costs of CNY ~62 million (~USD 8.7 million as of November 2025). Compared with a tier 2 risk assessment using a single fraction approach, this reduced the required remediation volume by 294,857 m^3 and saved ~324 million CNY (~USD 45.5 million as of November 2025) in costs.

Our carbon fraction proportion parameters for PHCs were primarily referenced from international research. We recommend further research on the compositional characteristics of PHCs in typical contamination types and regional contexts in China to refine risk assessments for PHC-contaminated sites.

We acknowledge several study limitations. First, our samples were primarily collected from a specific type of petrochemical site. Therefore, caution should be exercised if extrapolating these results to other industrial contexts or environmental media. Second, the mass fractions of sub-fractions used in the analysis were based on standards from Canada. Their direct application in regulatory frameworks of other regions requires careful consideration of applicability and potential need for localization. Despite these limitations, our findings provide decision support for the environmental regulation of PHCs C6–C9. The identified correlations can inform more efficient monitoring strategies and risk-based site management.

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Abbreviations

The following abbreviations were used in this manuscript:

PHC	Petroleum Hydrocarbon
TPH	Total Petroleum Hydrocarbon
BTEX	Benzene, Toluene, Ethylbenzene, and Xylene
USEPA	The United States Environmental Protection Agency
VOC	Volatile Organic Compound

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